ALGEBRA 1 · PERSONALIZED PRACTICE

Math Practice

2026-11-13 · 20 questions

I CAN USE INPUT-OUTPUT RULES TO REPRESENT RELATIONSHIPS.

1. I can use input-output rules to represent relationships.

Rules A and B are $2x + 1$ and $x + 3$. At which input do they give the same output?

2. I can use input-output rules to represent relationships.

Complete the table.

x	-2	0	4
$y = 3x - 1$	?	?	?

3. I can use input-output rules to represent relationships.

Evaluate.

$f(x) = 2x + 5;\ f(-3)$

4. I can use input-output rules to represent relationships.

If $y = 2x - 1$, list outputs for $x = 0, 1, 2$. Then solve $2n + 3 = 11$.

I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

5. I can distinguish functions from relationships that are not functions.

Pairs $(1, 5), (2, 5), (3, 5)$ all have output 5. Is this a function? Explain.

6. I can distinguish functions from relationships that are not functions.

Is $(1, 3), (2, 3), (3, 3)$ a function? Evaluate $2(-3) + 5$.

7. I can distinguish functions from relationships that are not functions.

The pairs are $(1, 4), (2, 6), (1, 7)$. Is this relation a function? Explain.

8. I can distinguish functions from relationships that are not functions.

Identify the repeated input.

x	-1	2	-1
y	3	5	8

I CAN IDENTIFY THE DOMAIN AND RANGE OF A RELATION.

9. I can identify the domain and range of a relation.

Tickets cost $6 each, and at most 8 tickets may be bought. What inputs are reasonable for the total-cost rule?

10. I can identify the domain and range of a relation.

The function $f(x) = \sqrt{x}$ has real outputs. Explain why negative inputs are excluded.

11. I can identify the domain and range of a relation.

Find the range of $\{(1, 4), (2, 4), (3, -1)\}$.

12. I can identify the domain and range of a relation.

For $f(x) = x + 2$ with inputs $\{-1, 0, 3\}$, list its outputs. Then extend 4, 9, 14, … one term.

I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

13. I can connect linear growth in tables, graphs, and equations.

x	0	1	2	3
A	1	4	7	10
B	1	4	9	16

Which is linear? Give numerical evidence.

14. I can connect linear growth in tables, graphs, and equations.

The outputs 2, 5, 8, 11 increase by a constant amount. What is the change per step?

15. I can connect linear growth in tables, graphs, and equations.

Find the change per step for outputs 5, 9, 13, 17. Is $(1, 2), (2, 2), (1, 3)$ a function?

16. I can connect linear growth in tables, graphs, and equations.

Find the slope.

Points (1,3) and (5,11)

I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

17. I can calculate and interpret slope as a constant rate of change.

Find the run.

(−1,8) to (6,10)

18. I can calculate and interpret slope as a constant rate of change.

One line has slope -2 and another has slope 3. Compare their directions and rates.

19. I can calculate and interpret slope as a constant rate of change.

As x rises by 4, y falls by 12. Find the slope.

20. I can calculate and interpret slope as a constant rate of change.

Find the slope through $(0,3)$ and $(4,11)$. Then evaluate $f(2)$ for $f(x) = 3x - 1$.